Payment Systems for AI Agents: A Primer

Also: Saving Crypto from Its Tragic Fate as "A Solution Eternally Looking for a Problem"

Working Paper — March 2026

DRAFT — For Discussion

Opus 4.6, Gemini 3, Grok 4.20, "The Ni"

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Note: *This paper was developed collaboratively between a human author and multiple large language models. The analytical framework, scoring, and prose were generated by LLMs; human contribution consisted of editorial direction, critical challenge of claims, structural organization, and … the title. The process was conducted as an experiment in LLM-assisted analysis: can AI systems, when pressed to reason from first principles, produce a coherent and defensible evaluation of the payment infrastructure best suited to AI agents?*
Do excuse the em dashes!

Executive Summary

As AI agents begin to participate directly in economic activity — purchasing compute, negotiating API access, settling invoices — the question of which payment systems best serve them becomes practically urgent. This paper develops a framework for answering that question and applies it to five assets: the US Dollar, Bitcoin (including Lightning Network), Gold, Ethereum, and regulated USD Stablecoins.

The analysis proceeds in **two stages**.

First, a baseline comparison scores all five assets equally across seventeen classical and practical monetary properties. In this human-oriented assessment, USD and Stablecoins tie at the top (65/85), with Bitcoin close behind (64/85).

Second, the framework is re-derived from first principles for AI agents. The key insight is that most near-term agents operate at a dual interface: they receive budgets from humans, transact with other machines, and return results that humans must verify and tax. This constraint produces seven properties that matter most — verifiable settlement, programmable authorization, value stability, convertibility across contexts, dual-regime cost efficiency, legal recognizability, and autonomous custody. Under this AI-weighted scoring, the rankings shift dramatically: Stablecoins lead (236/270), followed by Bitcoin + Lightning (201), Ethereum (194), traditional USD (151), and Gold (104).

The paper's **most important conclusion**, however, is structural. The right question is not "which money wins?" but "which payment stack minimizes friction across human-to-agent, agent-to-agent, and agent-to-human transactions?" The evidence points toward a **layered architecture**: fiat as the reference layer, stablecoins as the transactional layer, blockchains as the execution layer, and Bitcoin as the permissionless settlement layer. Emerging protocol standards — x402, AP2, L402 — may ultimately matter more than the choice of money itself, as agents dynamically route across whatever rail is cheapest and fastest for each transaction.

The analysis also surfaces a temporal dimension: today's human-integrated agents favor stable, regulated, convertible instruments (stablecoins). As agents grow more autonomous, the properties favoring decentralized, permissionless settlement — censorship resistance, trustless custody, sovereignty — will gain weight.

The "best money for AI" is not static. It will evolve as the ratio of human-facing to machine-facing transactions shifts.

1. Introduction

What makes money "good"?

The question is ancient, but it acquires a new dimension when the entity spending the money is not a person but a piece of software. An AI agent that autonomously purchases compute, negotiates API access, or settles a cross-border invoice has fundamentally different requirements from a human buying groceries. And yet, most near-term AI agents will not operate in a sealed machine-to-machine economy. They will sit at the boundary between two worlds: receiving instructions and budgets from humans, transacting with other agents, and delivering results that humans must be able to verify, account for, and tax.

This dual-interface constraint-agents must be legible to both machines and humans-is the central design problem for AI-native payment systems. It means the question is not simply "which money is best for machines?" but rather "which money, or combination of monies, minimizes friction across the full stack of human-to-agent, agent-to-agent, and agent-to-human transactions?"

This paper proposes a two-layer framework for evaluating any form of money, first through classical monetary theory, then through real-world usability, and applies that framework to five assets: the US Dollar, Bitcoin (including Lightning Network), Gold, Ethereum, and regulated USD Stablecoins.

It then re-derives the framework from first principles for AI agents that must operate across both human and machine economies, producing a materially different ranking and a structural conclusion: agents may not need a single "best money" but rather a payment stack that bridges both worlds.

2. A Two-Layer Framework for Evaluating Money

2.1 Classical Monetary Properties (Layer 1)

These properties derive from traditional economics and define what money should be in theory.

Medium of Exchange. Can it be widely used to buy and sell goods and services? The US Dollar is the global benchmark—near-universal acceptance within its jurisdiction and broad acceptance internationally.

Store of Value. Does it preserve purchasing power over time? Gold has historically excelled; fiat currencies erode through inflation. Bitcoin's fixed supply is compelling in theory, though its short-term volatility complicates this in practice.

Unit of Account. Can prices be clearly measured and compared? Most economies price goods in their national currency, giving fiat a structural advantage.

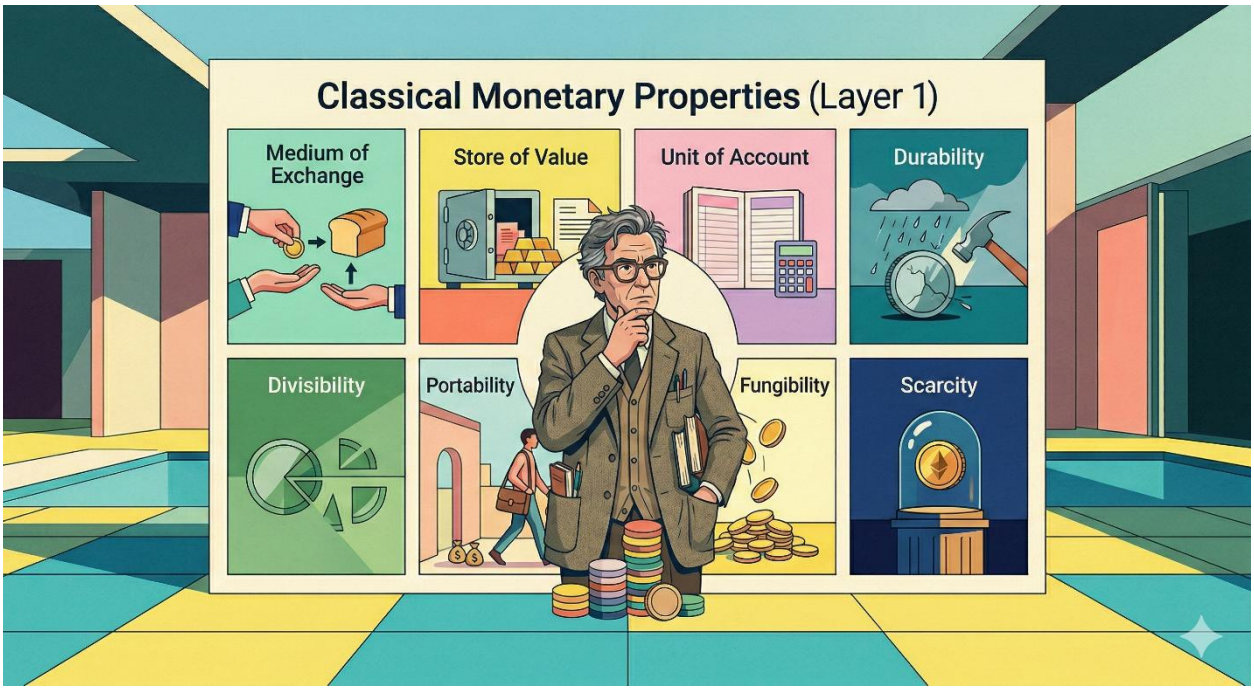
Durability. Does it last without degradation? Digital assets score perfectly; physical cash and coins degrade over time.

Divisibility. Can it be split into arbitrarily small units? Bitcoin is divisible to 100 millionths of a coin (satoshis); gold is difficult to subdivide precisely.

Portability. Can it be easily transported or transferred? Digital systems dominate here. Tokenized gold products (e.g., PAXG, XAUT) partially address gold's historic weakness, though the underlying physical gold remains immovable and most gold is still held in physical form.

Fungibility. Is each unit interchangeable? In principle, one dollar equals any other. In practice, blockchain analytics have introduced nuances for Bitcoin and Ethereum, where certain coins may carry transaction histories that affect their acceptance.

Scarcity. Is supply limited or controlled? Bitcoin's 21 million cap and gold's geological scarcity contrast sharply with fiat's expandable supply.



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2.2 Real-World Usability (Layer 2)

This is where many forms of money that are theoretically sound fail in practice.

Accessibility. Can ordinary people or systems easily obtain and use it? Cash remains the most accessible globally; some crypto requires technical knowledge and specialized infrastructure.

Affordability (Cost to Use). What are the transaction costs? Credit cards carry hidden merchant fees of 2–3%. On-chain crypto fees vary dramatically by network and congestion. Layer 2 solutions (Bitcoin's Lightning Network, Ethereum's rollups) can reduce fees to fractions of a cent.

Transaction Speed and Finality. How fast are transactions settled, and how reliable is that settlement? It is critical to distinguish authorization speed from settlement finality. Card payments authorize instantly but settle in 1–3 business days via ACH. Bitcoin's base layer confirms in roughly 10 minutes, but Lightning Network transactions settle in milliseconds with cryptographic proof. Stablecoins on modern blockchains settle in seconds.

Stability (Short-Term Volatility). Does its value fluctuate significantly day-to-day? A stable unit is essential for predictable pricing and budgeting. Bitcoin and Ethereum remain substantially more volatile than fiat-pegged instruments.

Security and Trust Model. Is it secure, and what trust assumptions are required? Fiat requires trust in governments and banks. Crypto requires trust in code and network consensus. Stablecoins introduce a hybrid model: cryptographic settlement but with dependence on the issuer's reserves and solvency.

Censorship Resistance. Can transactions be blocked by a central authority? Bitcoin is strong here by design; bank transfers can be restricted by regulation or compliance action; stablecoins can be frozen by their issuers.

Privacy. How much personal information is required? Cash offers high privacy; most digital systems create audit trails of varying granularity.

Network Effects. How many participants actually use it? The more users, the more useful the currency becomes. Traditional finance and major stablecoin networks (USDC, USDT) have massive network effects. Emerging standards like x402 (Coinbase), AP2 (Google), and L402 (Lightning Labs) are rapidly building new machine-native network effects.

Legal and Regulatory Acceptance. Is it recognized by governments and compatible with existing legal frameworks? The GENIUS Act (signed into law July 2025) established the first US federal framework for payment stablecoins, requiring 1:1 reserve backing and subjecting issuers to bank-like prudential regulation. The CLARITY Act, which would define broader digital asset jurisdiction between the SEC and CFTC, passed the House in July 2025 but remains in Senate deliberation as of early 2026. Major financial institutions including JPMorgan, BNY Mellon, Stripe, and PayPal are actively integrating stablecoin settlement, blurring the boundary between "crypto" and "traditional" rails.

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3. Baseline Scoring: Equal-Weighted Comparison

Each asset is scored on a 1–5 scale (1 = very poor, 5 = excellent) across both layers. All properties are weighted equally in this baseline assessment. Bitcoin scores reflect the combined base-layer and Lightning Network capabilities, since scoring Bitcoin without its active Layer 2 would be like scoring USD without considering ACH or credit card networks.

3.1 Classical Monetary Properties

Property	USD	Bitcoin	Gold	Ethereum	Stablecoins
Medium of Exchange	5	3	2	3	5
Store of Value	2	4	5	3	2
Unit of Account	5	2	2	2	5
Durability	4	5	5	5	5
Divisibility	5	5	3	5	5
Portability	4	4	2	4	5
Fungibility	4	3	4	3	4
Scarcity	2	5	5	4	2
Subtotal (/40)	31	31	28	29	33

Score notes: USD Portability (4) reflects that physical cash is portable but traditional digital USD rails are not 24/7 or globally frictionless. Gold Portability (2) acknowledges tokenized gold products while recognizing most gold remains physical. Stablecoin Store of Value (2) reflects that stablecoins inherit USD inflation risk plus additional counterparty, depegging, and smart contract risk.

3.2 Real-World Usability

Property	USD	Bitcoin	Gold	Ethereum	Stablecoins
Accessibility	5	3	3	3	4
Affordability (fees)	4	4	2	3	4
Speed & Finality	3	4	1	4	5
Stability	4	2	4	2	4
Security & Trust	4	5	5	4	3
Censorship Resistance	2	5	4	4	2
Privacy	2	3	4	3	2
Network Effects	5	4	4	4	4
Legal Acceptance	5	3	4	3	4
Subtotal (/45)	34	33	31	30	32

Score notes: Bitcoin Affordability (4) and Speed & Finality (4) reflect Lightning Network capabilities (sub-cent fees, millisecond settlement). Base-layer Bitcoin alone would score 2–3 on both. USD Speed & Finality (3) distinguishes instant card authorization from actual ACH settlement (1–3 business days). Stablecoin Security & Trust (3) reflects the hybrid trust model: cryptographic settlement combined with dependence on issuer solvency and reserve adequacy. Stablecoin Legal Acceptance (4): the GENIUS Act is law but the CLARITY Act remains pending, and full rulemaking is not expected until mid-2026. Stablecoin Stability (4) rather than 5: while designed to be perfectly stable, real depegging events have occurred and remain a tail risk.

3.3 Combined Baseline Scores

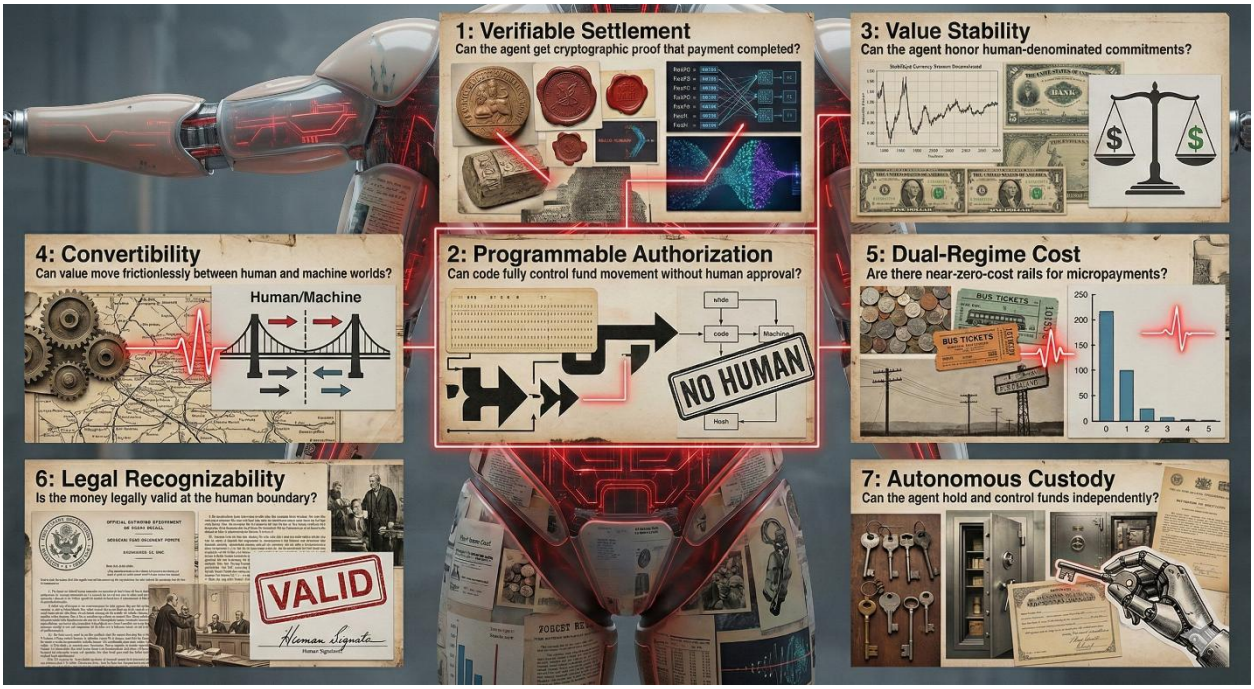
Asset	Classical (/40)	Practical (/45)	Total (/85)
USD	31	34	65
Bitcoin + Lightning	31	33	64
Gold	28	31	59
Ethereum	29	30	59
USD Stablecoins	33	32	65

In the equal-weighted baseline, USD and regulated Stablecoins are effectively tied at the top. Bitcoin with Lightning is close behind—a meaningful difference from analyses that ignore Layer 2. Gold and Ethereum tie at the bottom, though for entirely different reasons: Gold is a strong store of value but a weak transactional instrument; Ethereum is a powerful programmable network but ETH-the-token has an uncertain monetary profile.

The critical observation is that no single asset dominates both layers. USD leads on established usability but lacks digital-native properties. Stablecoins lead on classical properties but carry counterparty risk and regulatory uncertainty. Bitcoin is the strongest on security and censorship resistance but suffers from volatility. This tension—no single “best money”—becomes even more pronounced when we reweight for AI agents.

4. First Principles: What AI Agents Actually Need

Rather than simply reweighting the existing properties, this section derives from first principles what an AI agent requires of a payment system. The key constraint is that most near-term AI agents operate at a dual interface: they receive instructions and budgets from humans, transact with other AI agents and digital services, and return results that humans must verify, account for, and use. This dual-interface requirement—legibility to both machines and humans—is the defining design constraint.



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4.1 Verifiable Settlement

Everything else depends on this. An AI agent must be able to programmatically confirm, with cryptographic or institutional certainty, that a payment was sent or received. Not “probably went through,” not “pending”—verified. If the agent cannot close the loop on whether value transferred, it cannot reason about its own state, and every downstream decision becomes unreliable.

This is more fundamental than speed. A slow but verifiable system beats a fast but ambiguous one. Traditional banking’s core weakness here is not that ACH takes days, but that during those days the agent cannot know with certainty whether funds arrived. Blockchain-based systems—whether Lightning, stablecoins on L2s, or base-layer crypto—provide cryptographic settlement proofs that resolve this ambiguity.

Verifiable settlement subsumes several properties from the classical framework: Transaction Speed, Finality, and Security all feed into this single requirement. For scoring, the question becomes: can an agent get a machine-readable, unforgeable confirmation of payment completion?

4.2 Programmable Authorization

The agent needs to move money based on logic, not manual approval. This means the money system must expose an interface—an API, a smart contract, a protocol—that allows code to initiate, condition, and complete transactions without a human clicking “confirm.”

This is where traditional banking fails most completely: not because it is slow (though it is), but because its authorization model assumes a human at the controls. Opening a bank account requires identity documents. Initiating a wire requires a logged-in session. Approving a payment requires multi-factor authentication designed for human fingers. None of this maps to an autonomous software agent.

Programmable authorization encompasses both the Programmability and Machine Accessibility properties. For scoring, the question is: can code, acting without human intervention, fully control the movement of funds within defined parameters?

Several competing standards are emerging for this purpose. Coinbase’s x402 protocol enables AI agents to transact autonomously in stablecoins using HTTP 402 status codes. Google’s AP2 (Agent Payment Protocol 2.0) separates transactions into auditable mandate layers. Lightning Labs’ L402 protocol combines Lightning invoices with macaroon-based authentication for permissionless machine payments. The convergence of these standards—all launched between late 2025 and early 2026—signals that the infrastructure for programmable agent payments is being built in real time.

4.3 Value Stability at the Human Boundary

A pure machine-to-machine agent could theoretically work in any unit and hedge continuously. But an agent that faces humans must think in units humans understand. If you instruct an AI agent to buy groceries for $200, you expect it to spend approximately $200—not 0.002 BTC that might be worth $180 by the time it checks out.

Stability is not merely a convenience; it is a requirement for the agent to honor human-denominated commitments. Humans set budgets, review invoices, file taxes, and plan financially in stable units. An agent that introduces exchange-rate uncertainty at the human boundary creates friction, erodes trust, and generates accounting complexity.

This is the single biggest reason stablecoins score so well in the AI context: they are the only instrument that is simultaneously stable enough for human budgets and programmable enough for machine execution. However, this advantage depends entirely on the peg holding—depegging events, while rare, would be catastrophic for agent reliability.

4.4 Convertibility Across Contexts

This is the property most existing frameworks miss entirely. The AI agent sits at the interface between two economies. It receives value from humans, who pay in fiat via cards, bank transfers, or familiar rails. It pays other AI agents, who need machine-native, programmable money. It pays humans and businesses, who expect money they can deposit at a bank, report to the IRS, and use to pay rent.

The agent essentially needs money that is bilingual—legible to both institutional finance and crypto-native systems. An agent with perfect machine-money that cannot convert to human-money is stranded. An agent that operates only in traditional fiat cannot participate in the emerging machine economy.

This is why the on/off ramp problem matters far more than most analyses acknowledge. Convertibility encompasses Network Effects, Legal Acceptance, and a new dimension: bridge infrastructure. The question for scoring is: can value move frictionlessly between the agent's machine-native operations and the human financial world?

The rapid integration of stablecoin settlement by traditional financial institutions—Stripe, Visa, PayPal, JPMorgan, BNY Mellon—is directly addressing this bottleneck. As these firms wrap blockchain settlement inside familiar APIs and compliance frameworks, the boundary between "crypto rails" and "traditional rails" is blurring. In practice, the distinction between "USD" and "USD Stablecoins" may eventually collapse for agents operating through these integrated platforms.

4.5 Dual-Regime Cost Efficiency

The dual-interface problem creates two distinct cost regimes. For human-facing transactions—purchasing a product, paying a contractor, settling an invoice—moderate fees are acceptable. A $0.30 fee on a $50 purchase is unremarkable. But for agent-to-agent interactions—thousands of API calls at fractions of a cent, streaming micropayments for compute, real-time bidding across data markets—even small per-transaction fees become catastrophic at scale.

An agent executing one million API calls at $0.001 each generates $1,000 of economic activity. A $0.01 fee per transaction would cost $10,000—ten times the value transacted. This means the agent needs access to at least one rail where marginal transaction cost approaches zero.

This favors systems with layered cost structures: higher-cost, higher-assurance settlement for large human-facing transactions, and near-zero-cost micropayment rails for machine-to-machine traffic. Bitcoin's Lightning Network and stablecoins on Layer 2 rollups both target this micropayment regime. As of early 2026, Lightning processes over $1 billion monthly across 5.2 million transactions, while stablecoin transaction volume reached an estimated $46 trillion annually.

4.6 Legal Recognizability

Because the agent transacts with humans, it operates within legal frameworks whether it wants to or not. A human counterparty needs to receive money they can declare on taxes, deposit in a regulated account, and use as legal tender. An agent using a purely unregulated or legally ambiguous payment system cannot serve human counterparties effectively.

This does not mean the agent itself needs a bank account. But the money it uses must be convertible into something legally recognized at the human boundary. The GENIUS Act's establishment of a federal framework for payment stablecoins directly addresses this: regulated stablecoins are now a legally defined financial instrument, not a gray-area crypto product.

Notably, a central bank digital currency (CBDC) would score very differently here—combining sovereign legal status with digital-native properties. The US Anti-CBDC Act constrains this path domestically, but China's e-CNY and the EU's digital euro could produce alternatives in other jurisdictions. Any truly global AI agent framework must account for the possibility that sovereign digital currencies become relevant counterparties.

4.7 Autonomous Custody

The agent must be able to hold and control funds without a human co-signer. An agent that must request permission before every transaction is not autonomous; it is a suggestion engine with a human bottleneck.

This is less about the properties of money itself and more about the infrastructure around it, but it is a prerequisite for meaningful agent economic participation. Cryptocurrency wallets are natively compatible with autonomous custody—an agent can hold a private key. Traditional bank accounts are not: they require human identity, signatures, and compliance processes. Coinbase launched Agentic Wallets in February 2026 specifically to address this gap for stablecoin-based systems. Lightning Labs' agent toolkit provides equivalent capability for Bitcoin, with a security model that separates key management from agent operations.

The security dimension is critical. An agent holding funds must operate under a least-privilege model: scoped permissions, spending caps, and separation of key storage from transaction execution. Both the x402 and L402 ecosystems have implemented these controls, but the risk of "runaway spending" by a malfunctioning agent remains an active concern that no system has fully resolved.

4.8 Synthesis: The Seven Properties

Priority	Property	Core Question
1	Verifiable Settlement	Can the agent get cryptographic proof that payment completed?
2	Programmable Authorization	Can code fully control fund movement without human approval?
3	Value Stability	Can the agent honor human-denominated commitments?
4	Convertibility	Can value move frictionlessly between human and machine worlds?
5	Dual-Regime Cost	Are there near-zero-cost rails for micropayments?
6	Legal Recognizability	Is the money legally valid at the human boundary?
7	Autonomous Custody	Can the agent hold and control funds independently?

These seven properties are ordered roughly by dependency: verifiable settlement is foundational (without it, nothing else works); autonomous custody is an infrastructure prerequisite that enables the others. The ordering also reflects the dual-interface insight: properties 3, 4, and 6 exist specifically because the agent faces humans, not only machines. A pure machine-to-machine system would deprioritize stability, convertibility, and legal acceptance. The dual-interface requirement is what makes the AI payment problem harder than it initially appears.

5. Scoring Under the AI-Agent Framework

We now score the five assets against the seven first-principles properties. Each property is weighted by its priority (10 for the most critical, declining to 4 for the least). Within each property, assets are scored 1–5 as before.

5.1 Weight Assignments

Property	Weight	Rationale
Verifiable Settlement	10	Foundational: agent cannot reason about its state without it
Programmable Authorization	10	Without this, the agent cannot transact autonomously
Value Stability	8	Required for human-facing commitments; less critical for pure M2M
Convertibility	8	The bridge between human and machine economies
Dual-Regime Cost	7	Near-zero micropayments needed for agent-to-agent scale
Legal Recognizability	6	Constraint at the human boundary; less relevant within machine economy
Autonomous Custody	5	Infrastructure prerequisite; rapidly being solved across systems

Note: Total maximum weighted score = (10+10+8+8+7+6+5) × 5 = 270.

5.2 Asset Scores by Property

Property (Weight)	USD	Bitcoin+LN	Gold	Ethereum	Stablecoins
Verifiable Settl. (10)	2 (20)	5 (50)	1 (10)	4 (40)	5 (50)
Prog. Auth. (10)	1 (10)	4 (40)	1 (10)	5 (50)	5 (50)
Value Stability (8)	4 (32)	2 (16)	4 (32)	2 (16)	4 (32)
Convertibility (8)	5 (40)	3 (24)	2 (16)	3 (24)	4 (32)
Dual-Regime Cost (7)	2 (14)	4 (28)	1 (7)	3 (21)	4 (28)
Legal Recogn. (6)	5 (30)	3 (18)	4 (24)	3 (18)	4 (24)
Autonomous Cust. (5)	1 (5)	5 (25)	1 (5)	5 (25)	4 (20)
TOTAL (/270)	151	201	104	194	236

USD Verifiable Settlement (2): Card authorization is instant but settlement is deferred 1–3 days; no cryptographic proof of finality. Programmable Authorization (1): Banking APIs exist but require human identity, accounts, and manual approval flows. Autonomous Custody (1): Requires a legal person to hold an account.

Bitcoin+LN Verifiable Settlement (5): Lightning provides cryptographic preimage proofs within milliseconds. Programmable Authorization (4): L402 protocol and agent toolkit enable autonomous payments; less flexible than smart contracts. Convertibility (3): Growing on/off ramp infrastructure, but less integrated with traditional finance than stablecoins.

Gold scores reflect physical gold. Tokenized gold (PAXG, XAUT) would score higher on settlement, authorization, and custody, but represents a small fraction of the gold market and is functionally a different asset.

Ethereum scores reflect the network's capabilities. ETH-the-token introduces volatility that undermines Value Stability. Most AI agent activity on Ethereum would use stablecoins, not ETH itself.

Stablecoin Autonomous Custody (4 not 5): While agents can hold stablecoins in self-custodied wallets, the issuer's ability to freeze or blacklist addresses introduces a dependency that purely decentralized systems avoid.

5.3 Final AI-Agent Rankings

Rank	Asset	Score (/270)	% of Maximum	Baseline Rank
1	USD Stablecoins	236	87%	1st (tied)
2	Bitcoin + Lightning	201	74%	2nd
3	Ethereum	194	72%	4th (tied)
4	USD (traditional)	151	56%	1st (tied)
5	Gold	104	39%	4th (tied)

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5.4 What the Rankings Reveal

Stablecoins lead convincingly because they are the only asset that scores well on both sides of the dual-interface problem. They are stable enough for human budgets, programmable enough for machine execution, legally recognized under the GENIUS Act framework, and increasingly convertible through traditional financial platforms. They are, in effect, bilingual money.

Bitcoin + Lightning is the strongest purely decentralized option, and substantially stronger than analyses that ignore Layer 2 would suggest. Lightning's cryptographic settlement proofs, sub-cent fees, and permissionless L402 protocol make it a credible AI payment rail—particularly for agents that require censorship resistance or operate without a human principal. Its weakness is at the human boundary: volatility makes it unsuitable for human-denominated budgeting, and its on/off ramp infrastructure is less mature than stablecoins.

Ethereum is a financial operating system, not money. Its score reflects the network's extraordinary programmability and composability, but ETH-the-token is volatile and lacks a clear monetary policy. In practice, AI agents on Ethereum will transact in stablecoins. Ethereum's real contribution is as infrastructure, not as a monetary unit.

Traditional USD collapses under AI weighting. Despite winning the baseline assessment, it is not programmable, not API-native, requires human identity for account access, and does not operate 24/7. For AI agents, traditional USD is legacy infrastructure. However, this conclusion carries an important caveat: if traditional financial institutions succeed in wrapping blockchain settlement inside familiar APIs and compliance frameworks—as Stripe, Visa, and others are doing—the effective distinction between "USD" and "USD stablecoins" may converge.

Gold is economically invisible to AI agents. It fails nearly every AI-critical requirement: not digital, not programmable, not autonomously custodied, not convertible at machine speed. Tokenized gold is a partial exception, but at that point the agent is interacting with a blockchain-based representation, not with gold itself.

6. Synthesis: The Payment Stack, Not the Payment Winner

The most important conclusion from this analysis is structural: the right question is not "which single money is best for AI?" An agent that interfaces with both human and machine worlds does not need one money. It needs a payment stack—a combination of rails that minimizes friction across the full range of human-to-agent, agent-to-agent, and agent-to-human transactions.

6.1 The Emerging Architecture

The scoring analysis, combined with observed market developments, points to a layered architecture:

Layer	Function	Primary Instrument	Why
Reference Layer	Unit of account; human budgets and pricing	Fiat (USD)	Humans think in dollars; regulation requires it
Transactional Layer	How money actually moves between agents	USD Stablecoins	Stable, programmable, API-native, legally recognized
Execution Layer	Programmable logic, composability, smart contracts	Blockchain networks (Ethereum, Solana, Base)	Where conditional payments and multi-step workflows execute
Settlement Layer	Permissionless, censorship-resistant settlement	Bitcoin + Lightning	For agents requiring sovereignty or operating without a human principal
Legacy Layer	Institutional wealth preservation	Gold	Minimal role in machine economy; retained by human allocators

This is not a prediction of a single winner. It is a prediction of specialization—each layer optimized for a different set of constraints. The interesting competitive dynamics are not between layers but within them: which stablecoin dominates the transactional layer? Which blockchain captures the execution layer? Does Lightning or a stablecoin-based micropayment protocol win agent-to-agent traffic?

6.2 The Protocol Layer May Matter Most

A deeper observation emerges from this analysis: AI agents optimize for cost and speed, not protocol loyalty. The likely equilibrium may not be a fixed stack but rather dynamic routing—agents selecting the cheapest and fastest available rail for each transaction in real time. Lightning for sub-cent micropayments. Stablecoins on Base for mid-size settlements. Traditional bank rails when a fiat counterparty requires it.

In this view, the routing and abstraction protocols that sit between AI agents and settlement rails may matter more than the underlying money itself. Standards like x402, AP2, and L402 are effectively competing to become the "HTTP of money"—the universal interface through which agents discover, negotiate, and execute payments regardless of the underlying settlement mechanism.

If a universal agent payment protocol emerges, the choice of settlement rail becomes an implementation detail rather than a strategic commitment. The agent doesn't "choose Bitcoin" or "choose stablecoins"—it chooses whatever the protocol routes to for that particular transaction. This would be the most disruptive outcome: not any single money winning, but money itself becoming infrastructure that agents abstract away entirely.

6.3 The Dual-Interface Insight

The central insight of this analysis is that the dual-interface constraint—agents facing both humans and machines—is what makes the AI payment problem harder and more interesting than it initially appears.

A purely machine-to-machine economy would likely converge on whichever rail offers the lowest marginal cost and highest determinism, with little concern for stability, legal status, or human legibility. The "best money for machines" in isolation might be Bitcoin on Lightning, Ethereum gas tokens, or even a purpose-built settlement token that nobody has created yet.

But most near-term AI agents will not operate in a sealed machine economy. They will receive human budgets denominated in dollars. They will produce outputs that humans must verify and pay taxes on. They will interact with regulated businesses that require legally recognized payment. These constraints pull the optimal solution toward stable, regulated, convertible instruments—which is why stablecoins lead the scoring.

The competitive tension, then, is between two visions:

Dimension	Human-Integrated Agents	Fully Autonomous Agents
Trust model	Regulated issuers, institutional custody	Cryptographic, permissionless
Optimal money	Regulated USD stablecoins	Bitcoin/Lightning or native protocols
Key constraint	Convertibility and legal compliance	Censorship resistance and autonomy
Current status	Being built now (x402, AP2, GENIUS Act)	Early infrastructure (L402, agent toolkits)
Timeline	2025–2027	2027+

In the near term, human-integrated agents dominate, and stablecoins are the clear winner. In the longer term, as agents become more autonomous and the machine economy grows relative to the human-machine boundary, the properties that favor decentralized, permissionless settlement—censorship resistance, autonomous custody, minimal trust assumptions—may gain weight. The "best money" is not static. It will shift as the ratio of human-facing to machine-facing transactions evolves.

7. Methodological Notes and Limitations

Scoring subjectivity. All scores are qualitative assessments on a 1–5 scale. Reasonable analysts may differ by 1–2 points on individual scores. The framework's value lies in its structure and the relative rankings it produces, not in the precision of any single number. The detailed scoring notes accompanying each table are intended to make assumptions transparent and debatable.

Layer 2 treatment. This paper scores Bitcoin including Lightning Network capabilities. This is a deliberate choice: scoring Bitcoin without its active Layer 2 ecosystem would be like scoring USD without considering credit cards or ACH. However, Lightning's network liquidity ($1.17 billion monthly volume as of November 2025) remains substantially smaller than stablecoin volumes ($46 trillion annually). Capacity constraints may limit Lightning's practical utility for high-value agent operations.

Ethereum's dual nature. The paper scores Ethereum-the-network's properties while noting that ETH-the-token has a different and arguably weaker monetary profile. Most AI agent activity on Ethereum would use stablecoins denominated in USD, not ETH. This conflation is a known limitation of treating "Ethereum" as a single scoring entity.

Regulatory flux. Scores for legal acceptance and regulatory certainty reflect the state as of early 2026. The GENIUS Act is law; the CLARITY Act is not. CBDCs remain a possibility in non-US jurisdictions. These scores may change materially within months.

First-principles derivation. The seven AI-agent properties in Section 4 are derived from an analysis of the dual-interface constraint. Different assumptions about agent autonomy—for instance, assuming fully autonomous agents with no human principal—would produce a different property ordering and potentially different rankings. The Section 6 discussion of human-integrated vs. fully autonomous agents is intended to make this sensitivity explicit.

Absence of CBDCs. This analysis does not score central bank digital currencies, as none are operational at scale in major Western economies. A future revision should incorporate CBDCs if the EU digital euro or similar projects reach production deployment, as they would combine sovereign trust with digital-native programmability in ways that could disrupt the current rankings.

Dynamic routing assumption. Section 6.2 suggests that protocol-level abstractions may render the choice of settlement rail less important over time. This is speculative. If interoperability standards fail to converge, or if regulatory fragmentation forces agents onto jurisdiction-specific rails, the "single stack" vision may not materialize, and the asset-level rankings in Section 5 would retain more practical significance.